

AHMED MOHAMMED

Applied AI / ML Engineer | Computer Vision | M.Sc. Artificial Intelligence, JKU Linz
Linz, Austria | +43 681 20567353 | ahmed.mo.0595@gmail.com
linkedin.com/in/ahmed-3m | github.com/ahmed-3m | ahmed-3m.github.io



PROFESSIONAL SUMMARY

Machine learning engineer with an **M.Sc. in Artificial Intelligence from Johannes Kepler University Linz** (September 2026). My thesis, supervised by **Univ.-Prof. Dr. Sepp Hochreiter** and **graded 1**, uses conditional diffusion models as generative classifiers for out-of-distribution detection. A class-conditional separation loss I introduced cut the seed-to-seed standard deviation on **CIFAR-10** from 11.07 to 0.07 points while raising mean AUROC from 92.52% to **99.03%**. I applied the method to real inkjet inspection data in the PROFACTOR Zer0P project, and I build and run a cloud-hosted LLM assistant as a side project accepted into the tech2b incubator. My depth is Python, PyTorch and computer vision; I use AI-assisted development (Claude Code, GitHub Copilot) openly and daily. Available full time from **October 2026**.

CORE SKILLS

Deep Learning	PyTorch · PyTorch Lightning · Diffusion models (DDPM, UNet) · CNNs · Transformers · LSTM
Computer Vision	OOD detection · object detection (YOLOv8) · defect inspection · image classification · Roboflow
LLMs and GenAI	LLM API integration · multi-provider routing with escalation and failover · prompt engineering · structured output validation · retrieval with pgvector
Research Methods	Reproducible Hydra pipelines · ablation studies · AUROC and FPR95 · cross-validation · multi-seed runs
Cloud and Software	Python (advanced) · Git · Linux · Docker · REST APIs · GitHub Actions CI/CD · Supabase (PostgreSQL, row-level security, serverless functions) · TypeScript and React (familiar, built with AI-assisted tooling)
AI-Assisted Delivery	Claude Code (daily) · GitHub Copilot · Microsoft 365 Copilot · ChatGPT

RESEARCH AND PROFESSIONAL EXPERIENCE

M.Sc. Thesis Researcher, Machine Vision

Apr 2024 – Jul 2026

JKU Institute for Machine Learning with PROFACTOR GmbH

Industrial research project carried out in home office, coordinated remotely with the PROFACTOR engineering and experiment team

- Thesis: *Conditional Diffusion Models as Generative Classifiers for Out-of-Distribution Detection in Inkjet Print Quality Control*. Supervisor Univ.-Prof. Dr. Sepp Hochreiter, assistant supervisor Claus Hofmann, MSc. **Graded 1**.
- Introduced a **class-conditional separation loss** for conditional diffusion models. On **CIFAR-10** over three seeds it cut the seed-to-seed standard deviation from 11.07 to 0.07 points and raised mean OOD detection AUROC from 92.52% to **99.03%**; across five external benchmarks (CIFAR-100, Places365, FashionMNIST, SVHN, Textures) it reached 90.50% to 96.97% AUROC.
- Applied the framework to real industrial data in project **Zer0P** (Government of Upper Austria, zero-defect inkjet printing on building components): a two-stage YOLOv8 then conditional-diffusion pipeline for classifying experimental prints produced in a parameter-search feedback loop. Annotated the project dataset myself in Roboflow.
- Ran an eight-method comparison with five-fold cross-validation under one evaluation protocol, and reported where a supervised ResNet baseline stayed stronger on small, fine-grained inspection data.
- Built the experiment infrastructure in **PyTorch Lightning and Hydra**: seed-controlled runs, configuration tracking, and auditable raw-score artefacts so every reported number can be regenerated.

Solo Developer and Project Lead, Sihem (personal side project)

May 2026 – present

Accepted into the tech2b incubator | Linz, Upper Austria

- Design and run the full cloud stack behind an assistant for habits and daily routines, delivered as a Telegram bot and an installable web app: managed PostgreSQL with row-level security, serverless functions in Deno/TypeScript, scheduled jobs, secrets in a managed vault, and deployment through **GitHub Actions**.
- Model calls route across several LLM providers with escalation and failover, so one provider degrading does not take the service down and cost per request stays predictable. Wrote the incubator application, plan and pitch alone; web client (React and Vite) built with AI-assisted tooling.

AI Research Intern

Aug 2023 – Oct 2023

Karunya University | Coimbatore, India

- Built deep learning models for **EEG-based motor imagery classification** in brain-computer interface systems: a decoding problem on noisy, non-stationary, multi-channel time series.
- Designed signal preprocessing and feature-extraction pipelines and compared CNN and RNN architectures for real-time decoding. Written up as a paper on the portfolio site.

Technical Staff, Electronics

2018 – Oct 2020

Ledzone | Karaköy, Istanbul, Turkey

- Hands-on electronics work in a small workshop and parts business: PCB assembly and soldering, LED and motor driver circuit build-up, reading schematics and PCB layouts, and functional test, fault finding and final inspection with multimeter, oscilloscope and function generator.

EDUCATION

M.Sc. Artificial Intelligence

Johannes Kepler University (JKU) Linz, Austria | Oct 2020 – Sep 2026

Graduated September 2026. Thesis graded 1. Supervisor Univ.-Prof. Dr. Sepp Hochreiter, assistant supervisor Claus Hofmann, MSc. Coursework: Deep Learning and Neural Nets I and II, LSTM and Recurrent Neural Nets, Computer Vision, Probabilistic Models, Explainable AI, Reinforcement Learning, Deep Reinforcement Learning, Introduction to Robotic Systems, Control Systems.

B.Sc. Mechatronics Engineering

Eastern Mediterranean University, Famagusta, Cyprus | Feb 2015 – Jan 2018

Graduation project: SCARA robotic system for dynamic object tracking (kinematics, sensor integration, real-time control loop, mechanical and electrical integration of the arm).

Electrical Engineering studies (no degree)

Higher Technological Institute, Cairo, Egypt | Oct 2011 – Jan 2015

Transferred into the Mechatronics Engineering programme at Eastern Mediterranean University.

REFEREES

Univ.-Prof. Dr. Sepp Hochreiter (thesis supervisor, JKU Linz) and Dr. Christian Eitzinger (Head of Machine Vision, PROFACTOR GmbH). Contact details available on request.

LANGUAGES

Arabic Native

English B2, professional working

German B1 CEFR, preparing for B2

WORK AUTHORIZATION AND AVAILABILITY

Resident in Linz. I graduated from JKU Linz in September 2026 and am eligible now for the **Rot-Weiß-Rot Karte für Studienabsolventen** (Red-White-Red Card for graduates): **no AMS labour market test, no quota**, and no agency or sponsorship cost for the employer, who only confirms the position and salary. Available to start full time from **October 2026**.